

1.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum (x_i - \bar{x}) =$ 0	$\sum (x_i - \bar{x})^2 =$ 10	$\sum (y_i - \bar{y}) =$ 0	$\sum (x_i - \bar{x})(y_i - \bar{y}) =$ 8

a. $\bar{x} = 1, \bar{y} = 2$

b. $b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2 \bar{x} = 2 - 1 \times 0.8 = 1.2$

c. $\sum x_i^2 = 9 + 4 + 1 + 1 = 15$

$\sum x_i y_i = 12 + 4 + 3 - 1 = 18$

$\sum (x_i - \bar{x})^2 = 10$

$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \times 1^2 = 10 = \sum (x_i - \bar{x})^2$

$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

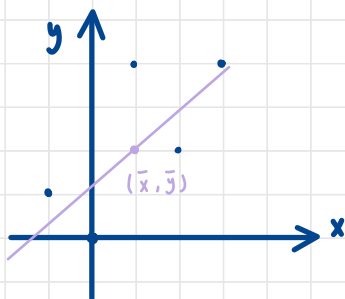
$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum (x_i - \bar{x})(y_i - \bar{y})$

d. $\hat{y}_i = 1.2 + 0.8 x_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$ 10	$\sum y_i =$ 10	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i \hat{e}_i =$ 0

e.

f.



fitting line will always
pass through $(\bar{x}, \bar{y})$

$$g. \quad \bar{y} = 2 \quad , \quad b_1 + b_2 \bar{x} = 1.2 + 0.8 \times 1 = 2 = \bar{y}$$

$$h. \quad \bar{\hat{y}} = \frac{\sum_{i=1}^5 (b_1 + b_2 x_i)}{5} = \frac{10}{5} = 2 = \bar{y}$$

$$i. \quad \sum \hat{e}_i^2 = 0.4^2 + (-0.8)^2 + 1^2 + 0.6^2 + (-1.2)^2 = 3$$

$$\hat{\sigma}^2 = \frac{3.6}{5-2} = 1.2$$

$$se(b_2) = \sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}} = \sqrt{\frac{1.2}{10}} \approx 0.3464$$

22. a.

Call:

lm(formula = total^yscore ~ small, data = star5)

Residuals:

Min	1Q	Median	3Q	Max
-283.20	-50.94	-7.20	42.80	334.80

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	918.201	1.155	795.00	< 2e-16 ***
small	13.741	2.107	6.52	7.61e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.48 on 5784 degrees of freedom
Multiple R-squared: 0.007297, Adjusted R-squared: 0.007125
F-statistic: 42.51 on 1 and 5784 DF, p-value: 7.609e-11

$$\hat{y} = 918.201 + 13.741 \times \text{small}$$

是小班 = 1, 不是小班 = 0

⇒ 是小班制比不是的總分平均起來多 13.741 分

b.

Call:
lm(formula = readscore ~ small, data = star5)

Residuals:

Min	1Q	Median	3Q	Max
-120.089	-22.089	-4.089	15.911	191.911

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	435.0887	0.4969	875.539	< 2e-16 ***
small	5.4631	0.9067	6.025	1.79e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.62 on 5784 degrees of freedom
Multiple R-squared: 0.006237, Adjusted R-squared: 0.006066
F-statistic: 36.3 on 1 and 5784 DF, p-value: 1.793e-09

$$\hat{y} = 435.0887 + 5.4631 \times \text{small}$$

⇒ 是小班制比不是的
read分平均起來多
5.4631分

Call:
lm(formula = mathscore ~ small, data = star5)

Residuals:

Min	1Q	Median	3Q	Max
-163.113	-34.113	-5.113	29.887	142.887

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.1126	0.7473	646.468	< 2e-16 ***
small	8.2775	1.3635	6.071	1.36e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.55 on 5784 degrees of freedom
Multiple R-squared: 0.006331, Adjusted R-squared: 0.006159
F-statistic: 36.85 on 1 and 5784 DF, p-value: 1.355e-09

$$\hat{y} = 483.1126 + 8.2775 \times \text{small}$$

⇒ 是小班制比不是的
math分平均起來多
8.2775分

c.

Call:
lm(formula = totalscore ~ aide, data = star5)

Residuals:

Min	1Q	Median	3Q	Max
-289.50	-51.36	-7.36	42.50	334.64

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	924.497	1.205	767.503	< 2e-16 ***
aide	-6.140	2.027	-3.029	0.00247 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.69 on 5784 degrees of freedom
Multiple R-squared: 0.001584, Adjusted R-squared: 0.001411
F-statistic: 9.174 on 1 and 5784 DF, p-value: 0.002466

$$\hat{y} = 924.497 - 6.14 \text{ aide}$$

⇒ 有另一個隨班人員和總成績呈負相關, 有的比沒有的少-6.14分

d.

```
Call:
lm(formula = readscore ~ aide, data = star5)

Residuals:
    Min       1Q   Median       3Q      Max
-122.435  -21.438   -4.435   15.565  191.562

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  437.4347    0.5182   844.19  <2e-16 ***
aide        -1.9966     0.8720   -2.29   0.0221 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.7 on 5784 degrees of freedom
Multiple R-squared:  0.0009055, Adjusted R-squared:  0.0007328
F-statistic: 5.242 on 1 and 5784 DF, p-value: 0.02208
```

$$\hat{y} = 437.4347 - 1.9966 \text{ aide}$$

有通班人員比沒有的read成績
低 1.9966

```
Call:
lm(formula = mathscore ~ aide, data = star5)

Residuals:
    Min       1Q   Median       3Q      Max
-167.062  -33.062   -3.062   30.081  143.081

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  487.062    0.779   625.271  < 2e-16 ***
aide        -4.143     1.311   -3.161   0.00158 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.66 on 5784 degrees of freedom
Multiple R-squared:  0.001724, Adjusted R-squared:  0.001551
F-statistic: 9.989 on 1 and 5784 DF, p-value: 0.001583
```

$$\hat{y} = 487.062 - 4.143 \text{ aide}$$

有通班人員比沒有的math成績
低 4.143

25.

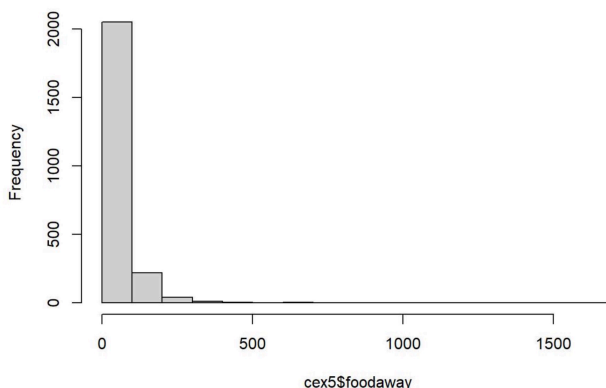
a. Min. 1st Qu. Median Mean 3rd Qu. Max.

 0.00 14.12 33.15 51.45 69.44 1645.56

mean = 51.45 , median = 33.15

25th percentiles = 14.12 , 75th percentiles = 69.44

Histogram of FOODAWAY



b.

```
# advanced
summary(cex5$foodaway[cex5$advanced == 1])

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00  21.67   48.34   75.37  89.86 1179.00

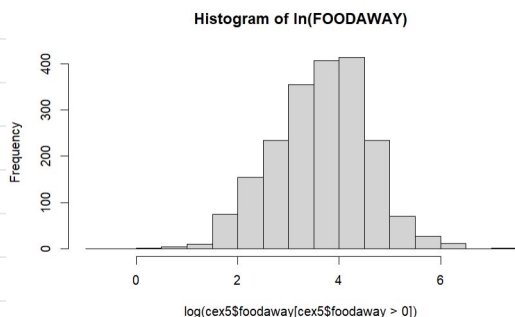
# college
summary(cex5$foodaway[cex5$college == 1])

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00  14.90   36.11   54.90  72.22 1645.56

# no advanced or college
summary(cex5$foodaway[cex5$advanced == 0 & cex5$college == 0])

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   9.63   25.02   38.32  55.97  437.78
```

c.



$\ln(\text{foodway})$, foodway 观测值数量不同, 原因是数量.

$f(x) = \ln(x)$, only when $x > 0$ 有定义, when $x = 0$, $\ln x \rightarrow \infty$

d.

```
## Call:
## lm(formula = log(foodaway) ~ income, data = cex5[cex5$foodaway :
##      0, ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.3034 -0.6003  0.0654  0.6119  3.6262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.1364599   0.0420567   74.58  <2e-16 ***
## income       0.0069283   0.0004898   14.15  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.898 on 2003 degrees of freedom
## Multiple R-squared:  0.09083,    Adjusted R-squared:  0.09038
## F-statistic: 200.1 on 1 and 2003 DF,  p-value: < 2.2e-16
```

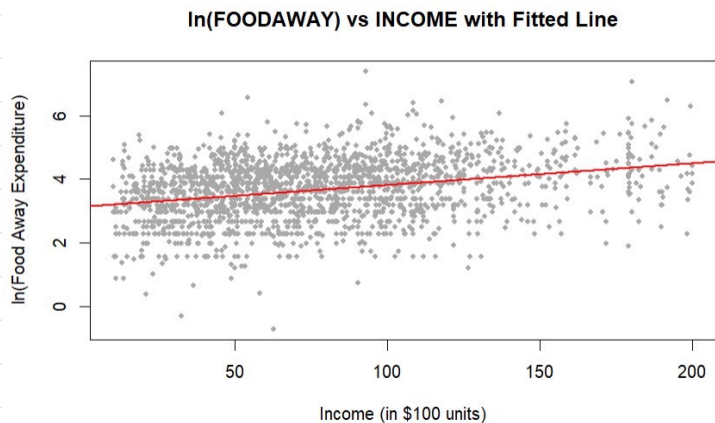
$$\ln(\text{foodway}) = 3.1365 + 0.0069 \text{ income}^x$$

$$\text{slope} = \frac{dy}{dx} = \beta_2 \exp(\beta_1 + \beta_2 x)$$

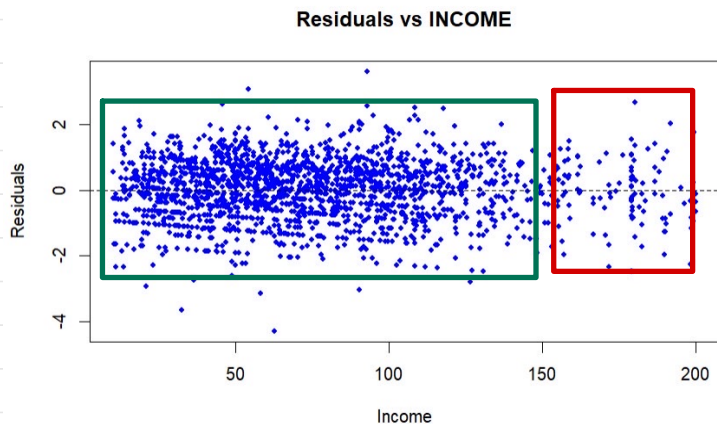
$$= 0.062 (3.1365 + 0.0069 \text{ income})$$

income 增加 1 单位, 预期额外支出会增加 slope 1倍

e.



f.



綠色區域及紅色區域有些不太均勻,但情況輕微
表示可能迴歸模型的假設會有問題